Board 2892: Perturbations by the 2022 Hunga-Tonga Volcano Eruption in the MLT Region Investigated Using the WACCM-X Simulation and Meteor Radar Observations

Wednesday, 13 December 2023

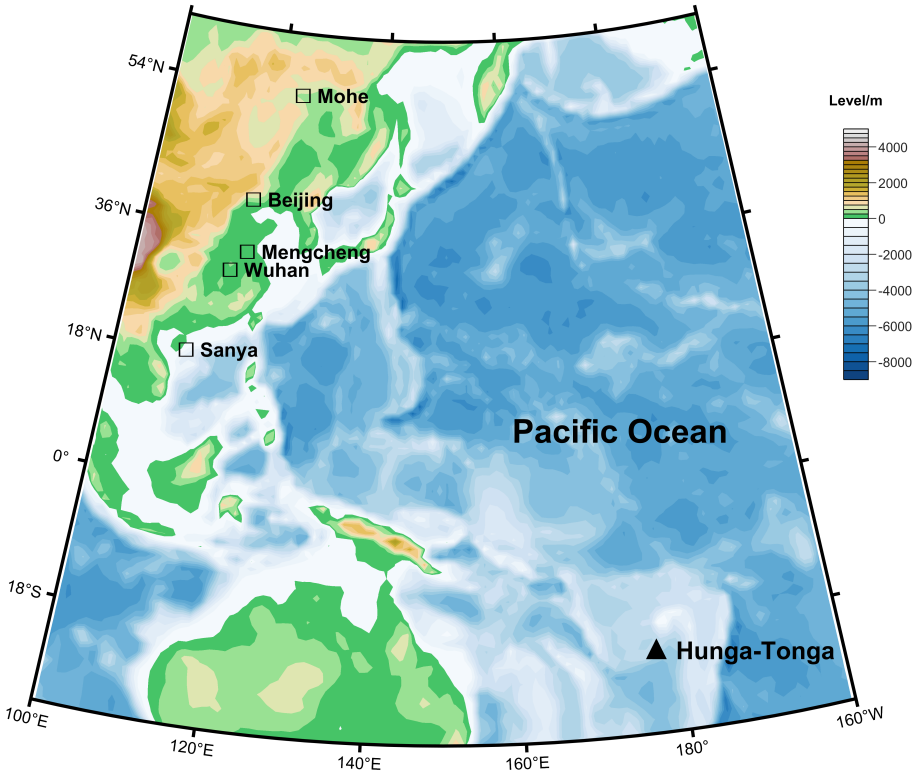
16:10 - 20:30

Poster Hall A-C - South (Exhibition Level, South, Moscone Center)

Abstract

The High-resolution Whole Atmosphere Community Climate Model with thermosphere/ionosphere extension (WACCM-X) and meteor radars at different latitudes are utilized to simulate and observe the atmospheric responses to the eruption of the Hunga-Tonga Volcano on January 15, 2022. Specifically, the meteor radars located at Mohe (53.5°N, 122.3°E), Beijing (40.3°N, 116.2°E), Mengcheng (33.3°N, 116.5°E), Wuhan (30.5°N, 114.2°E), and Sanya (18.3°N, 109.6°E) are used. The simulation results are compared to the ground-based meteor radar observations regarding perturbations of wind, neutral density, and temperature in the Mesosphere and Lower Thermosphere (MLT) region. By combining simulations and observations, we aim to elucidate the dynamic mechanism underlying the MLT region's atmospheric perturbations triggered by the eruption of the Hunga-Tonga

Volcano Eruption.



Plain-language Summary

We used sophisticated computer models and specialized instruments to study how the atmosphere reacts to the eruption of the Hunga-Tonga Volcano on January 15, 2022. We analyzed data from multiple meteor radars in China. Our goal was to understand the changes in wind patterns, density, and temperature in the upper atmosphere caused by the volcanic eruption. By combining these observations with computer simulations, we uncovered the reasons behind these atmospheric disruptions. This research provides insights into the impact of volcanic eruptions on our atmosphere, testing weather and climate models, and helps us prepare for future events.

Ask a question or comment on this session (not intended for technical support questions).

Have a question or comment? Enter it here.

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